

MTH00050 – Combinatorial Mathematics

Lecture 2: General counting methods for arrangements and selections

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(Tentative) Schedule 2026

No.	Date	Location	Topic
1	13/01 16/01	C24 C24/E302	Introduction, Sets, Proofs
2	20/01 23/01	C24 C24/E302	General counting methods
3	27/01 30/01	C24 C24/E302	Inclusion-exclusion principle
4	03/02 06/02	C24 C24/E302	Recurrence relations
5	03/03 06/03	C24 C24/E302	Generating functions
			Mid-term test

No.	Date	Location	Topic
6	10/03 13/03	C24 C24/E302	Graphs (Part I)
7	17/03 20/03	C24 C24/E302	Graphs (Part II)
8	24/03 27/03	C24 C24/E302	Path problems
9	31/03 03/04	C24 C24/E302	Tree problems
10	07/04 10/04	C24 C24/E302	Network flows (Part I)
11	14/04 17/04	C24 C24/E302	Network flows (Part II)
			Final exam

Course topics

0. Introduction
1. Set and counting
2. General counting methods for arrangements and selections
3. Inclusion-exclusion principle and Mobius inversion
4. Generating functions
5. Recurrence relations
6. Graphs
7. Tree problems
8. Path problems
9. Network flows

Outline

1. Simple arrangements and selections
2. Arrangements and selections with repetitions
3. Distributions
4. Binomial identities

Simple arrangements and selections

Definition 2. A permutation of n distinct objects is an arrangement, or ordering, of the n distinct.

Simple arrangements and selections

Definition 2. A **permutation** of n distinct objects is an arrangement, or ordering, of the n distinct.

Definition 3. An **r -permutation** of n distinct objects is an arrangement using r of the n distinct.

Simple arrangements and selections

Definition 4. A **r -combination** of n distinct objects is an **unordered selection**, or subset, of r out of the n objects.

We use $P(n, r), C(n, r) := C_n^r := \binom{n}{r}$ to denote the number of r -permutations and r -combinations, respectively, of a set of n objects.

Simple arrangements and selections

From the multiplication principle, we obtain

$$P(n, 2) = n(n - 1),$$

$$P(n, 3) = n(n - 1)(n - 2),$$

$$P(n, n) = n(n - 1)(n - 2) \dots 1 = n!$$

and

$$P(n, r) = n(n - 1)(n - 2) \dots [n - (r - 1)] = \frac{n!}{(n - r)!}$$

Simple arrangements and selections

All **r -permutations** of n objects can be generated by first picking any r -combination of the n objects and the arranging these r objects in any order. Thus:

$$P(n, r) = C(n, r) \times P(r, r),$$

and solving for $C(n, r)$, we have:

$$\binom{n}{r} = C(n, r) = \frac{P(n, r)}{P(r, r)} = \frac{n!/(n-r)!}{r!} = \frac{n!}{r!(n-r)!}$$

Example 12: Ranking Wizards

How many ways are there to rank n candidates for the job of chief wizard? If the ranking is made at random (each ranking is equally likely), what is the probability that the *fifth candidate, Gandalf*, is in *second place*?



Example 12: Ranking Wizards

Solution. A ranking is simply an arrangement, or permutation, of the n candidates.

- For the 1st position, there are n choices.
- For the 2nd position, there are $n - 1$ choices.
- For the 3rd position, there are $n - 2$ choices.
- And so on.

The total number of arrangements is:

$$n \times (n - 1) \times (n - 2) \times \cdots \times 1 = n!$$

Thus, the number of ways to rank n candidates is $n!$ ways.

Example 12: Ranking Wizards

Solution. We need to find the probability that the fifth candidate, Gandalf, is ranked in the second position when the ranking is made at random (each ranking is equally likely):

$$\text{Prob}(\text{Gandalf second}) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}.$$

As we know that the number of ways to rank n candidates is $n!$ ways. Now, we need to count outcomes where Gandalf is ranked second.

Let n candidates as $C_1, C_2, \dots, C_n$ where Gandalf is C_5 .

Example 12: Ranking Wizards

Solution. To count the number of favorable rankings:

- Fix Gandalf in the second position
- Arrange the remaining candidates

So, the number of favorable outcomes (rankings where Gandalf is in second place) is: $(n - 1)!$

Thus, the probability that Gandalf is in second place is:

$$\text{Prob}(\text{Gandalf second}) = \frac{(n - 1)!}{n!} = \frac{1}{n}.$$

Example 13: Arrangements with Repeated Letters

How many ways are there to arrange the seven letters in the word SYSTEMS? In how many of these arrangements do the three Ss appear consecutively?

Example 13: Arrangements with Repeated Letters

Solution. The word "SYSTEMS" has seven letters: S, Y, S, T, E, M, S where:

- The letter S appears 3 times.
- The letters Y, T, E, and M each appear 1 time.

We first pick the positions that the other four letters E, M, T, Y will occupy in the seven-letter arrangement, and then the three Ss will fill the remaining three positions in one way. There are seven possible positions for E, six for M, five for T, and four for Y.

Thus, there are $P(7, 4) = 7!/3! = 840$ arrangements.

Example 13: Arrangements with Repeated Letters

To find the number of arrangements where the three S's appear consecutively, we can treat the three S's as a single "super letter" or block. The word "SYSTEMS" now consists of:

- 1 block of [SSS]
- The remaining letters: Y, T, E, M (each appearing once)

So, the problem reduces to arranging the five distinct letters, Y, T, E, M and SSS (treated as a single letter), which can be done in $5! = 120$ ways.

Example 14: Poker Probabilities

- How many 5-card hands (subsets) can be formed from a standard 52-card deck?
- If a 5-card hand is chosen at random, what is the probability of obtaining a flush (all five cards in the hand are in the same suit)?
- What is the probability of obtaining three, but not four, Aces?



Example 14: Poker Probabilities

- a) A 5-card hand is a subset of five cards chosen from the 52 cards in a deck, and so there are:

$$C(52, 5) = \binom{52}{5} = \frac{52!}{47! 5!} = 2,598,960.$$

Example 14: Poker Probabilities

- b) To find the probability of a flush, we need to find the number of 5-card subsets with all cards of the same suit. There are four suits, and a subset of five cards from the 13 cards in a given suit can be chosen in $C(13, 5) = 13!/(5! 8!) = 1287$ ways. So there are $4 \times 1287 = 5148$ flushes. Thus, we have:

$$\text{Prob}(5 - \text{card hand is a flush}) = \frac{5148}{2,598,960} = 0.00198$$

 POKERSTARS POKER HAND RANKINGS					
▲ BEST		1	ROYAL FLUSH		
		2	STRAIGHT FLUSH		
		3	FOUR OF A KIND		
		4	FULL HOUSE		
		5	FLUSH		
WORST ▼		6	STRAIGHT		
		7	THREE OF A KIND		
		8	TWO PAIR		
		9	ONE PAIR		
		10	HIGH CARD		

Example 14: Poker Probabilities

- c) To count the number of hands with exactly three Aces, we must pick three of the four Aces – done in $C(4,3) = 4$ ways – and then fill out the hand with two card chosen from the 48 non-Ace cards – done in $C(48,2) = 1128$ ways. So, there are $4 \times 1128 = 4512$ hands with exactly three Aces. Finally, we get:

$$\text{Prob}(5 - \text{card hand has exactly three Aces}) = \frac{4512}{2,598,960} = 0.00174.$$

The Set Composition Principle

- Enumerate a set using the multiplication principle by multiplying the number of ways to form a **first part** by the number of ways to form a **second part** (for a given first part).
- The first and second parts must be disjoint—elements in the set must be uniquely identifiable as belonging to either the first or second part.

Example 13 (continued): Arrangements with Repetitions

How many arrangements of the seven letters in the word SYSTEMS have the E occurring somewhere before the M? How many arrangements have the E somewhere before the M and the three Ss grouped consecutively?

Example 13 (continued): Arrangements with Repetitions

Solution. Count arrangements of the 7 letters (S, Y, S, T, E, M, S) where E appears before M.

- Choose 2 positions for E and M: $C(7, 2) = 21$ ways.
- Place E in the earlier position, M in the later (1 way).
- Arrange Y, T in 2 of the 5 remaining positions: $P(5, 2) = 20$ ways.
- Place 3 S's in the 3 remaining position: 1 ways

Thus, the answer is: $21 \times 1 \times 20 \times 1 = 420$.

Example 13 (continued): Arrangements with Repetitions

Solution. Count arrangements where E is before M and the 3 S's are consecutive.

- Treat 3 S's as one "super letter" [SSS], reducing to 5 units: Y, T, E, M, [SSS].
- Choose 2 positions for E and M: $C(5,2) = 10$ ways.
- Place E in the earlier position, M in the later: 1 way
- Arrange Y, T, [SSS] in the 3 remaining positions: $3! = 6$ ways.

Thus, the answer is: $10 \times 1 \times 6 = 60$.

Example 15: Counting Defective Products

A manufacturing plant produces ovens. At the last stage, an inspector marks the ovens A (acceptable) or U (unacceptable). How many different sequences of 15 A s and U s are possible in which the third U appears as the twelfth letter in the sequence?

Example 15: Counting Defective Products

Solution. The sequence has 15 positions, and each position can be either A or U. Without any constraints, the total number of possible sequences would be 2^{15} since there are 2 choices (A or U) for each of the 15 positions.

- If the third U appears at the twelfth letter in the sequence, then the subsequence composed of the first 11 letters must have exactly two Us (and nine As). So there are $C(11, 2) = 55$ possible sequences for the first 11 letters.

Example 15: Counting Defective Products

- There is one way to pick the twelfth letter—it is specified to be U.
- The remaining three letters in the sequence can be either A or U, which is $2^3 = 8$ ways.

All together, the number of sequences is:

$$55 \times 1 \times 8 = 440.$$

Example 16: Probability of Repeated Digits

What is the probability that a 4-digit campus telephone number has one or more repeated digits?

Hint. There are $10^4 = 10,000$ different 4-digit phone numbers, so we break the problem into four different cases of repetitions:

- a) All four digits are the same.
- b) three digits are the same, the other is different.
- c) two digits are the same, the other two digits are also the same (e.g., 2828).
- d) two digits are the same, the other two digits are each different (e.g., 5105).

Outline

1. Simple arrangements and selections
2. Arrangements and selections with repetitions
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Example 17: Arrangements of *banana*

How many arrangements are there of the six letters b, a, n, a, n, a?

A possible arrangement can be illustrated as:

n	a	b	n	a	a
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Example 17: Arrangements of *banana*

How many arrangements are there of the six letters b, a, n, a, n, a?

A possible arrangement can be illustrated as:

n	a	b	n	a	a
---	---	---	---	---	---

⇒ The key is to focus on the subset of positions where the *as* go and the subset of positions where the *ns* go.

Example 17: Arrangements of *banana*



Solution.

- First choosing the subset of 3 positions in the arrangement where the *as* will go: $\binom{6}{3} = 20$ ways.
- Then the subset of two positions (out of the remaining three) where the *ns* will go: $\binom{3}{2} = 3$ ways.
- Finally, the last remaining position gets the *b*: $\binom{1}{1} = 1$ way

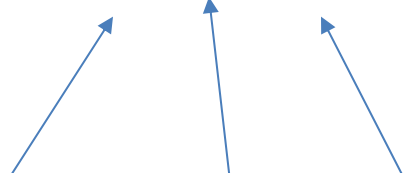
Thus, we have: $\binom{6}{3} \times \binom{3}{2} \times \binom{1}{1} = 20 \times 3 \times 1 = 60$ ways.

Example 17: Arrangements of *banana*

1	2	3	4	5	6

Solution.

$$\frac{6!}{3! 2! 1!} = 60$$



 a n b

Theorem 1. If there are n objects, with r_1 of type 1, r_2 of type 2, ..., and r_m of type m , where $r_1 + r_2 + \cdots + r_m = n$, then the number of arrangements of these n objects, denoted as $P(n; r_1, r_2, \dots, r_m)$ is

$$P(n; r_1, r_2, \dots, r_m) = \binom{n}{r_1} \binom{n - r_1}{r_2} \cdots \binom{n - r_1 - r_2 - \cdots - r_{m-1}}{r_m} = \frac{n!}{r_1! r_2! \cdots r_m!}.$$

Example 18: Ordering Hot Dogs

How many different ways are there to select six hot dogs from three varieties of hot dog?

3 Loại Hot Dog Có Sẵn

Classic



Hot dog truyền thống

Cheese



Hot dog phô mai

Veggie



Hot dog chay

Example 18: Ordering Hot Dogs

Solution. Suppose the three varieties are **classic**, **cheese** and **veggie**. Let a selection be written down on an order form in the following fashion:

classic	cheese	veggie
x	xxxx	x

Each x represents a hot dog. The request shown on the form above is **one classic**, **four cheese**, and **one veggie**. Since the hot dog chef knows that the sequence of dogs on the form is **classic**, **cheese** and **veggie**, the request can simply be written as xxxxx|x| without column headings.

Example 18: Ordering Hot Dogs

We observe that:

- Any selection of r hot dogs will consist of some sequence of r x s and two $|$ s.
- Conversely, any sequence of r x s and two $|$ s represents a unique selection: **the x s before the first $|$ count the number of classic dogs; the x s between the two $|$ s count cheese; and the final x s count veggie.**

⇒ There is a one-to-one correspondence between orders and such sequences!.

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⇒ There is a one-to-one correspondence between orders and such sequences!.

Counting the number of sequences of six x s and two $|$ s is an arrangement-with-

repetition problem whose answer is $P(8; 6, 2) = \binom{8}{6} \binom{2}{2} = \frac{8!}{6!2!} = 28$

Theorem 2. The number of selections with repetition of r objects chosen from n types of objects is:

$$\binom{n + r - 1}{r - 1}.$$

Example 19: Grouping Classes

Nine students, **three from Ms. A's class**, **three from Mr. B's class**, and **three from Ms. C's class**, have bought a block of nine seats for their school's homecoming game. If three seats are randomly selected for each class from the nine seats in a row, what is the probability that the **three A students**, **three B students**, and **three C students** will each get a block of three consecutive seats?

Example 19: Grouping Classes

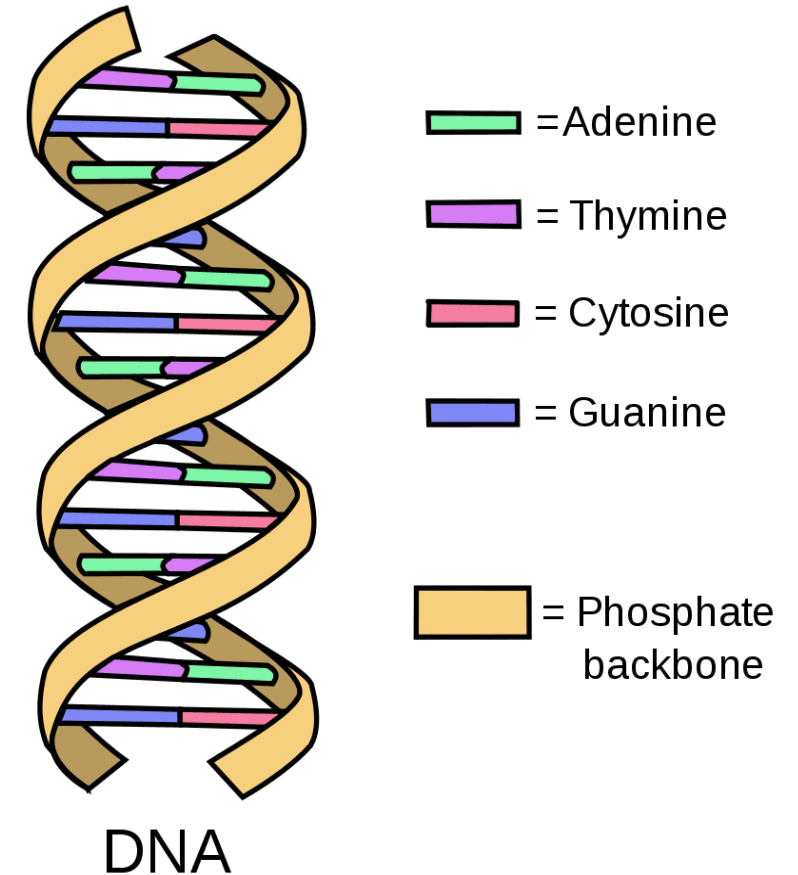
Solution.

- The number of arrangements three As, three Bs, and three Cs in the row of nine seats is: $P(9; 3,3,3) = 9!/3! 3! 3! = 1680$ ways.
- If the three students of each class are to sit together, then we want to count outcomes that are arrangements of the three blocks, AAA, BBB, and CCC. Thus, instead of nine letters, we really are working now with three composite letters. There are $3! = 6$ ways to arrange these three composite letters.

Thus, the probability that each class sits together is $6 / 1680$.

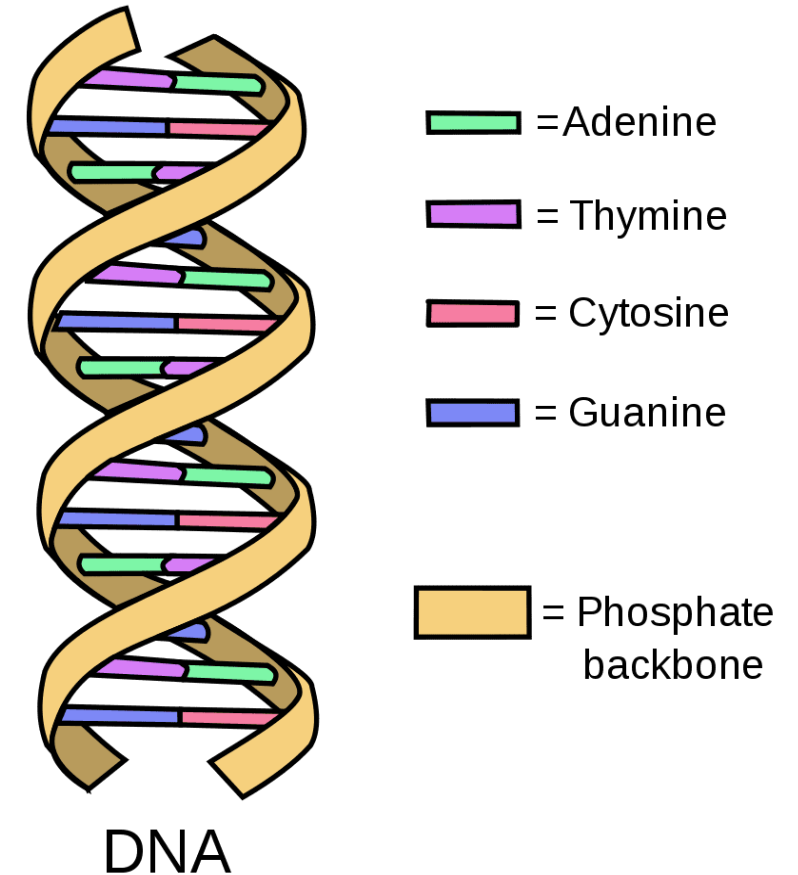
Example 20: Sequencing Genes (1/)

The genetic code of organisms is stored in DNA molecules as a long string of four nucleotides: **A** (adenine), **C** (cytosine), **G** (guanine), and **T** (thymine). Short strings of DNA can be “sequenced”—the sequence of letters determined—by various modern biotech methods.



Example 20: Sequencing Genes (2/)

Although the DNA sequence for a single gene typically has hundreds or thousands of letters, *there exist special enzymes* that will split a long string into short fragments by *breaking the string immediately following each appearance of a particular letter*.



Example 20: Sequencing Genes (3/

Suppose a **C**-enzyme (which splits after each appearance of C) breaks a 20-letter string into eight fragments, which are identified to be: **AC**, **AC**, **AAATC**, **C**, **C**, **C**, **TATA**, **TGGC**. Note that each fragment, except the last one on the string, must end with a **C**. How many different strings could have given rise to this set of fragments?

Example 20: Sequencing Genes (4/)

- Since the fragment TATA does not end with a C, it must go at the end of the string.

.....TATA

- The other seven fragments can occur in any order.
- These fragments consist of **three** Cs, **two** ACs, **one** AAATC, and **one** TGGC.
- Treat each fragment as a letter. Then we must arrange **seven** letters, **three** of one type, **two** of a second type, and **one** each of two other types.
- There are $P(7; 3, 2, 1, 1) = 420$ possible arrangements of the fragments to form a 20-letter string

Example 21: Sequences with Varying Repetitions

How many ways are there to form a sequence of 10 letters from four *a*s, four *b*s, four *c*s, and four *d*s if each letter must appear at least twice?

Example 21: Sequences with Varying Repetitions

Remark. To apply Theorem 1 we need to know exactly how many as, bs, cs, and ds will be in the arrangement. Thus we have to break this problem into a set of subproblems:

- *First category:* Four appearances of one letter and two appearances of each other letter.
- *Second category:* Three appearances of two letters and two appearances of the other two letters.

Example 21: Sequences with Varying Repetitions

In the first category:

- There are four cases for choosing which letter occurs four times.
- To arrange four of one letter and two of the three others, we have:

$$P(10; 4, 2, 2, 2) = 18,900 \text{ ways.}$$

Example 21: Sequences with Varying Repetitions

In the second category:

- For choosing which two of the four letters occur three times:

$$C(4,2) = 6 \text{ cases.}$$

- To arrange three of two letters and two of the two others, we have:

$$P(10; 3,3,2,2) = 25,200 \text{ ways.}$$

Thus, we have $4 \times P(10; 4,2,2,2) + 6 \times P(10; 3,3,2,2)$ ways.

Example 22: Selecting Doughnuts

How many ways are there to fill a box with a dozen doughnuts chosen from five different varieties with the requirement that at least one doughnut of each variety is picked?

Solution. Left as an exercise to the students.

Example 23: Selections with Lower and Upper Bounds

How many ways are there to pick a collection of exactly 10 balls from a pile of red balls, blue balls, and purple balls if there must be at least five red balls? If at most five red balls?

Solution. Left as an exercise to the students.

Outline

1. Simple arrangements and selections
2. Arrangements and selections with repetitions
3. **Distributions**
4. Binomial identities

Distributions

- Generally, a distribution problem is equivalent to an arrangement or selection problem with repetition.
- Specialized distribution problems must be broken up into subcases that can be counted in terms of simple permutations and combinations (with and without repetition)

Distributions

- Generally, a distribution problem is equivalent to an arrangement or selection problem with repetition.
- Specialized distribution problems must be broken up into subcases that can be counted in terms of simple permutations and combinations (with and without repetition)

Distributions of distinct objects $\Leftrightarrow$ arrangements

and

Distributions of identical objects $\Leftrightarrow$ selections

Basic Models for Distributions

Distinct Objects. The process of distributing r distinct objects into n different boxes is equivalent to putting the distinct objects in a row and stamping one of the n different box names on each object. Thus, there are

$$n \times n \times \cdots \times n \text{ (} r \text{ ns)} = n^r$$

distributions of the r distinct objects.

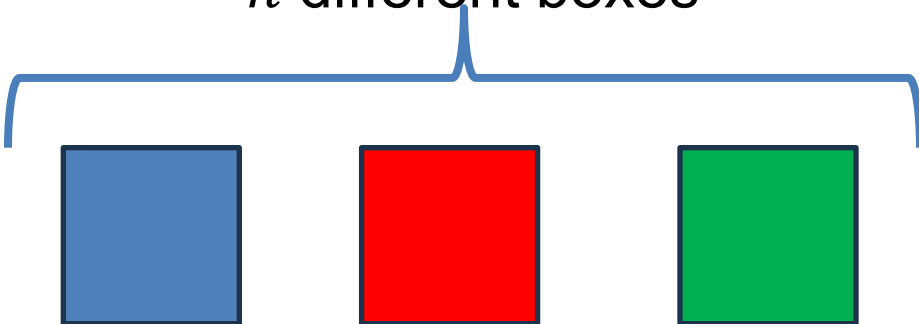
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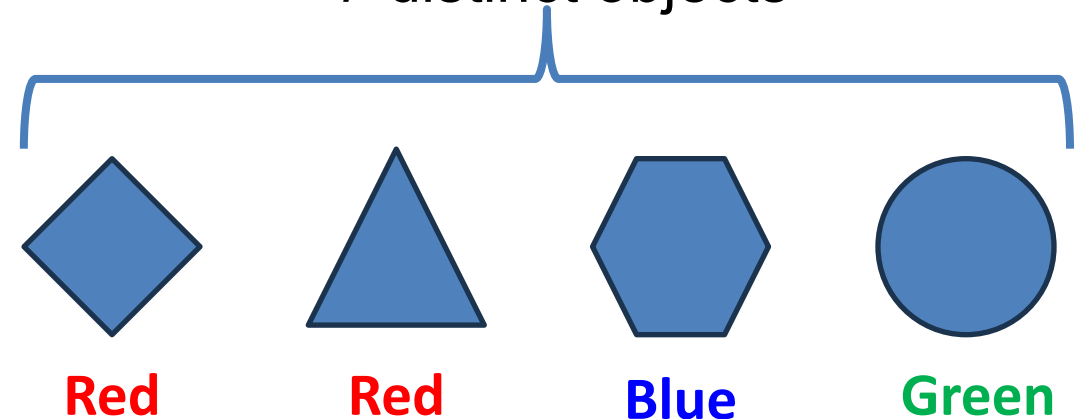
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n different boxes

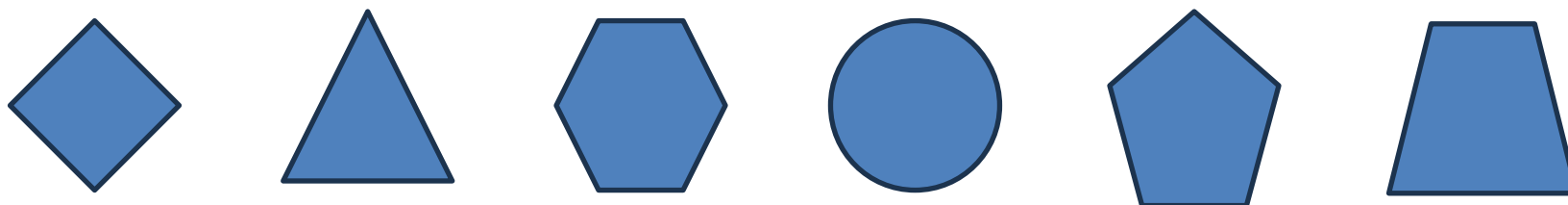


r distinct objects



Specified amount in each box

If r_i objects must go in box i , $1 \leq i \leq n$, then there are $P(r; r_1, r_2, \dots, r_n)$ distributions.



6 distinct objects



How many in each box?

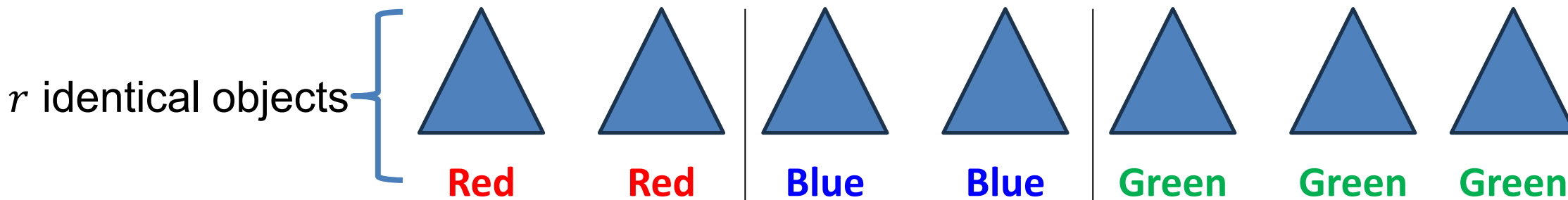
$$\binom{6}{3} \binom{6-3}{1} \binom{6-3-1}{2} = P(6; 3, 1, 2)$$

Basic Models for Distributions

Identical Objects. The process of distributing r identical objects into n different boxes is equivalent to choosing an (unordered) subset of r box names with repetition from among the n choices of boxes. Thus, there are

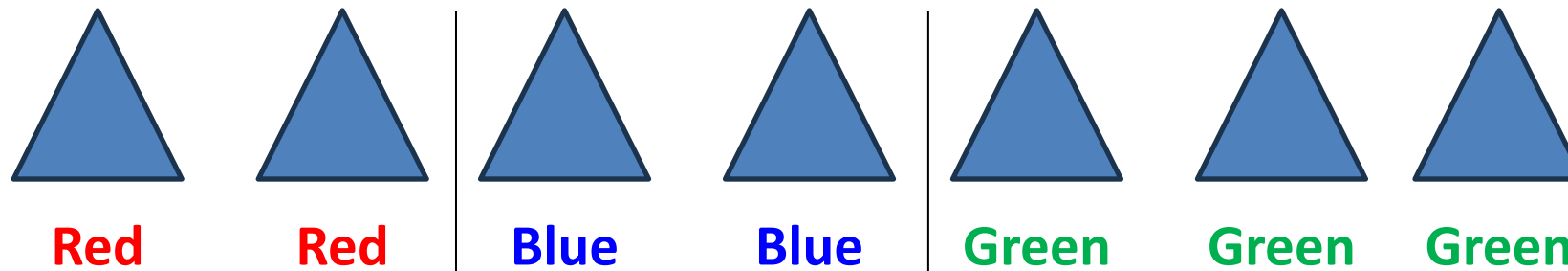
$$\binom{r + n - 1}{r} = \frac{(r + n - 1)!}{r! (n - 1)!}$$

distributions of the r identical objects.



Basic Models for Distributions

Example: 7 identical objects into 3 distinct boxes



$$7 = 2 + 2 + 3$$

Example 24: Assigning Diplomats

How many ways are there to assign 100 different diplomats to five different continents? How many ways if 20 diplomats must be assigned to each continent?

Example 24: Assigning Diplomats

How many ways are there to assign 100 different diplomats to five different continents? How many ways if 20 diplomats must be assigned to each continent?

Solution. According to the model for distributions of distinct objects, this assignment equals the number of sequences of length 100 involving the five continental destination, 5^{100} .

Example 24: Assigning Diplomats

How many ways are there to assign 100 different diplomats to five different continents? **How many ways if 20 diplomats must be assigned to each continent?**

Solution. The constraint that 20 diplomats go to each continent means that each continent name should appear 20 times in the sequence. Thus, we have: $P(100; 20, 20, 20, 20, 20) = 100! / (20!)^5$ ways.

Example 25: Bridge Hands

In bridge, the 52 cards of a standard card deck are randomly dealt 13 apiece to players North, East, South, and West. What is the probability that West has all 13 spades? That each player has one Ace?

Example 25: Bridge Hands

In bridge, the 52 cards of a standard card deck are randomly dealt 13 apiece to players North, East, South, and West. What is the probability that West has all 13 spades? That each player has one Ace?

Solution.

- Count the ways West can get all spades: 1 way.
- Count the ways to distribute the $52 - 13 = 39$ non-spade cards among the 3 other hands: $P(39; 13, 13, 13)$ ways

Example 25: Bridge Hands

In bridge, the 52 cards of a standard card deck are randomly dealt 13 apiece to players North, East, South, and West. What is the probability that West has all 13 spades? That each player has one Ace?

Solution. So, the probability what West has all the spades is:

$$\frac{39!}{(13!)^3} / \frac{52!}{(13!)^4} = 1 / \frac{52!}{13! 39!} = 1 / \binom{52}{13}.$$

Example 25: Bridge Hands

In bridge, the **52 cards** of a standard card deck are randomly dealt **13 apiece** to players North, East, South, and West. What is the probability that West has all 13 spades? **That each player has one Ace?**

Solution.

- First distribute the Aces: $4!$ ways.
- Then, distribute the 48 non-Aces: $P(48; 12, 12, 12, 12) = 48! / (12!)^4$ ways.

Example 25: Bridge Hands

In bridge, the **52 cards** of a standard card deck are randomly dealt **13 apiece** to players North, East, South, and West. What is the probability that West has all 13 spades? **That each player has one Ace?**

Solution. So, the probability that each player gets an Ace is:

$$\frac{4! P(48; 12, 12, 12, 12)}{P(52; 13, 13, 13, 13)} = \frac{4! 48!}{(12!)^4} \bigg/ \frac{52!}{(13!)^4} = \frac{(13!)^4}{(12!)^4} \times \frac{4! 48!}{52!} = 13^4 / \binom{52}{4}$$

$$= 0.105$$

Example 26: Distributing Candy

How many ways are there to distribute 20 (identical) sticks of red licorice and 15 (identical) sticks of black licorice among five children?

Example 26: Distributing Candy

How many ways are there to distribute 20 (identical) sticks of red licorice and 15 (identical) sticks of black licorice among five children?

Solution. Left as an exercise to the students.

Example 27: Distributing Balls

Show that the number of ways to distribute r identical balls into n distinct boxes with at least one ball in each is $C(r - 1, n - 1)$. With at least r_1 balls in the first box, at least r_2 balls in the second box, ..., and at least r_n balls in the n -th box, the number is $C(r - r_1 - r_2 - \cdots - r_n + n - 1, n - 1)$.

Example 27: Distributing Balls

Solution.

- First put one ball in each of the r boxes.
- Then count the ways to distribute without restriction the remaining $r - n$ balls into the n boxes:

$$C((r - n) + n - 1, (r - n)) = \frac{[(r - n) + n - 1]!}{(r - n)! (n - 1)!} = C(r - 1, n - 1)$$

Example 27: Distributing Balls

Solution.

- First, for each I , put r_i balls in the i -th box.
- There are now $r - r_1 - r_2 - \cdots - r_n$ balls left.
- Finally, count the ways to distribute without restriction the remaining $r - r_1 - r_2 - \cdots - r_n$ balls into the n boxes.
- So this can be done in

$$\begin{aligned} & C((r - r_1 - r_2 - \cdots - r_n) + n - 1, (r - r_1 - r_2 - \cdots - r_n)) \\ &= C((r - r_1 - r_2 - \cdots - r_n) + n - 1, n - 1) \end{aligned}$$

Example 28: Integer Solutions

How many non-negative integer solutions are there to the equation

$$x_1 + x_2 + x_3 + x_4 = 12$$

with:

a) $x_i \geq 0$?

b) $x_i \geq 1$?

c) $x_1 \geq 2, x_2 \geq 2, x_3 \geq 4, x_4 \geq 0$?

Example 28: Integer Solutions

Solution.

a) $x_i \geq 0$

An example of one solution is: $x_1 = 2, x_2 = 3, x_3 = 3, x_4 = 4$.

Two ways to think about this:

1. Let x_i be the number of (identical) objects in box i , or
2. Let x_i be the number of objects of type i chosen.

The number of integer solutions is: $C(12 + 4 - 1, 12) = 455$ ways.

Example 28: Integer Solutions

b) $x_i \geq 1$?

There are: $C(12 - 4 + (4 - 1), 4 - 1) = C(8 + 4 - 1, 4 - 1) = 165$ ways.

c) $x_1 \geq 2, x_2 \geq 2, x_3 \geq 4, x_4 \geq 0$?

There are: $C(12 - (2 + 2 + 4) + (4 - 1), 4 - 1) = C(4 + 4 - 1, 4 - 1) = 35$ ways.

Equivalent forms for selection with repetition

Equivalent Forms for Selection with Repetition.

(**r hot dogs and n varieties**)

1. The number of ways to select r objects with repetition from n different types of objects.
2. The number of ways to distribute r identical objects into n distinct boxes.
3. The number of nonnegative integer solutions

$$x_1 + x_2 + \cdots + x_n = r.$$

Example 28: Ingredients for a Witch's Brew

A warlock goes to a store with \$5 to buy ingredients for his wife's Witch's Brew. The store sells bat tails for 25c/ apiece, lizard claws for 25c/ apiece, newt eyes for 25c/ apiece, and calf blood for \$1 a pint bottle. How many different purchases (subsets) of ingredients will \$5 buy?

Solution. Left as an exercise to the students.

Example 29: Binary Patterns

What fraction of binary sequences of length 10 consists of a (positive) number of 1s, followed by a number of 0s, followed by a number of 1s, followed by a number of 0s? An example of such a sequence is 1110111000

Solution. Left as an exercise to the students.

Summarization

Ways to Arrange, Select, or Distribute r Objects from n Items or into n Boxes

Arrangement (Ordered Outcome) or Distribution of Distinct Objects

Combination (Unordered Outcome) or Distribution of Identical Objects

	<i>Arrangement (Ordered Outcome) or Distribution of Distinct Objects</i>	<i>Combination (Unordered Outcome) or Distribution of Identical Objects</i>
No repetition	$P(n, r)$	$C(n, r)$
Unlimited repetition	n^r	$C(n + r - 1, r)$
Restricted repetition	$P(n; r_1, r_2, \dots, r_m)$	

Outline

1. Simple arrangements and selections
2. Arrangements and selections with repetitions
3. Distributions
4. **Binomial identities**

Table of binomial coefficients

- Compute a table of binomial coefficients using

$$\binom{n}{k} = \frac{n!}{k! (n - k)!}.$$

- We'll look at several patterns. First, the nonzero entries of each row are symmetric; e.g., row $n = 4$ is

$$\binom{4}{0}, \binom{4}{1}, \binom{4}{2}, \binom{4}{3}, \binom{4}{4} = 1, 4, 6, 4, 1$$

which reads the same in reverse. **Conjecture:** $\binom{n}{k} = \binom{n}{n - k}$.

Table of binomial coefficients

$\binom{n}{k}$	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
$k = 0$	1	0	0	0	0	0	0
$k = 1$	1	1	0	0	0	0	0
$k = 2$	1	2	1	0	0	0	0
$k = 3$	1	3	3	1	0	0	0
$k = 4$	1	4	6	4	1	0	0
$k = 5$	1	5	10	10	5	1	0
$k = 6$	1	6	15	20	15	6	1

A binomial coefficient identity

Theorem 3. For non-negative integers $k \leq n$,

$$\binom{n}{k} = \binom{n}{n-k}, \quad \text{including } \binom{n}{0} = \binom{n}{n} = 1$$

A binomial coefficient identity

Theorem 3. For nonnegative integers $k \leq n$,

$$\binom{n}{k} = \binom{n}{n-k}, \quad \text{including } \binom{n}{0} = \binom{n}{n} = 1$$

Proof.

$$\binom{n}{k} = \frac{n!}{k! (n-k)!} = \frac{n!}{(n-k)! k!} = \binom{n}{n-k}.$$

Table of binomial coefficients

$\binom{n}{k}$	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$	Total
$k = 0$	1	0	0	0	0	0	0	1
$k = 1$	1	1	0	0	0	0	0	2
$k = 2$	1	2	1	0	0	0	0	4
$k = 3$	1	3	3	1	0	0	0	8
$k = 4$	1	4	6	4	1	0	0	16
$k = 5$	1	5	10	10	5	1	0	32
$k = 6$	1	6	15	20	15	6	1	64

Sum of binomial coefficients

Theorem 4. For integers $n \geq 0$, we have: $\sum_{k=0}^n \binom{n}{k} = 2^n$

Sum of binomial coefficients

Theorem 4. For integers $n \geq 0$, we have: $\sum_{k=0}^n \binom{n}{k} = 2^n$

Proof. Based on the Binomial Theorem

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k},$$

plug in $x = y = 1$.

Recursion for binomial coefficients

- A recursion involves solving a problem in terms of smaller instances of the same type of problem.
- Example: Consider 3-element subsets of [5]:

Subsets without 5	Subsets with 5
{1, 2, 3}	{1, 2, 5}
{1, 2, 4}	{1, 3, 5}
{1, 3, 4}	{1, 4, 5}
{2, 3, 4}	{2, 3, 5}
	{2, 4, 5}
	{3, 4, 5}

Recursion for binomial coefficients

Theorem 5. For nonnegative integers n, k :

$$\binom{n+1}{k+1} = \binom{n}{k} + \binom{n}{k+1}$$

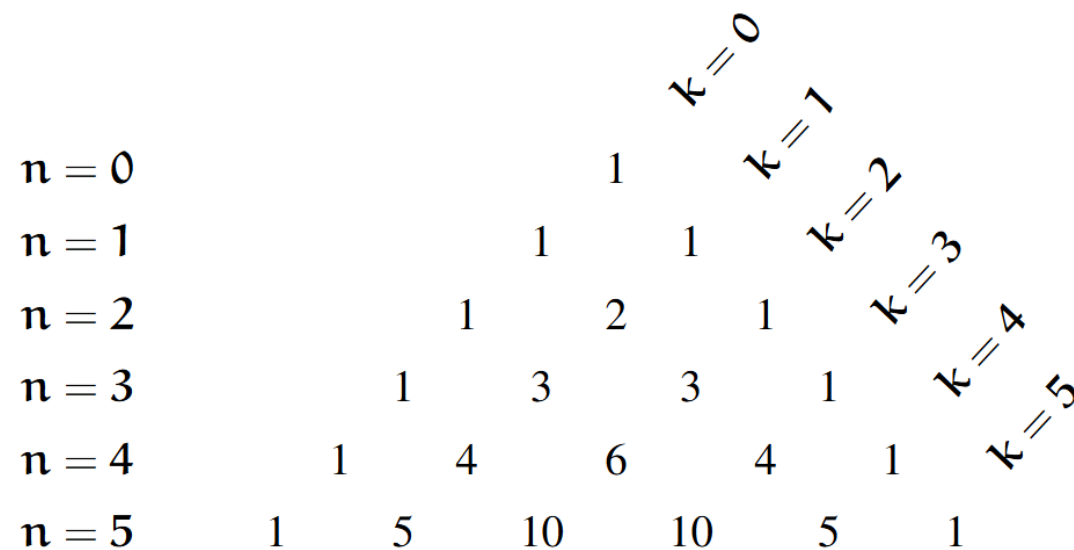
- The initial conditions are

$$\binom{n}{0} = 1, \forall n \geq 0, \quad \binom{0}{k} = 0, \forall k \geq 1$$

- For $n \geq 0$, the only 0-element subset of $[n]$ is $\emptyset$, so $\binom{n}{0} = 1$.
- For $k \geq 1$, there are no k -element subset of $[n] = \emptyset$, so $\binom{0}{k} = 0$

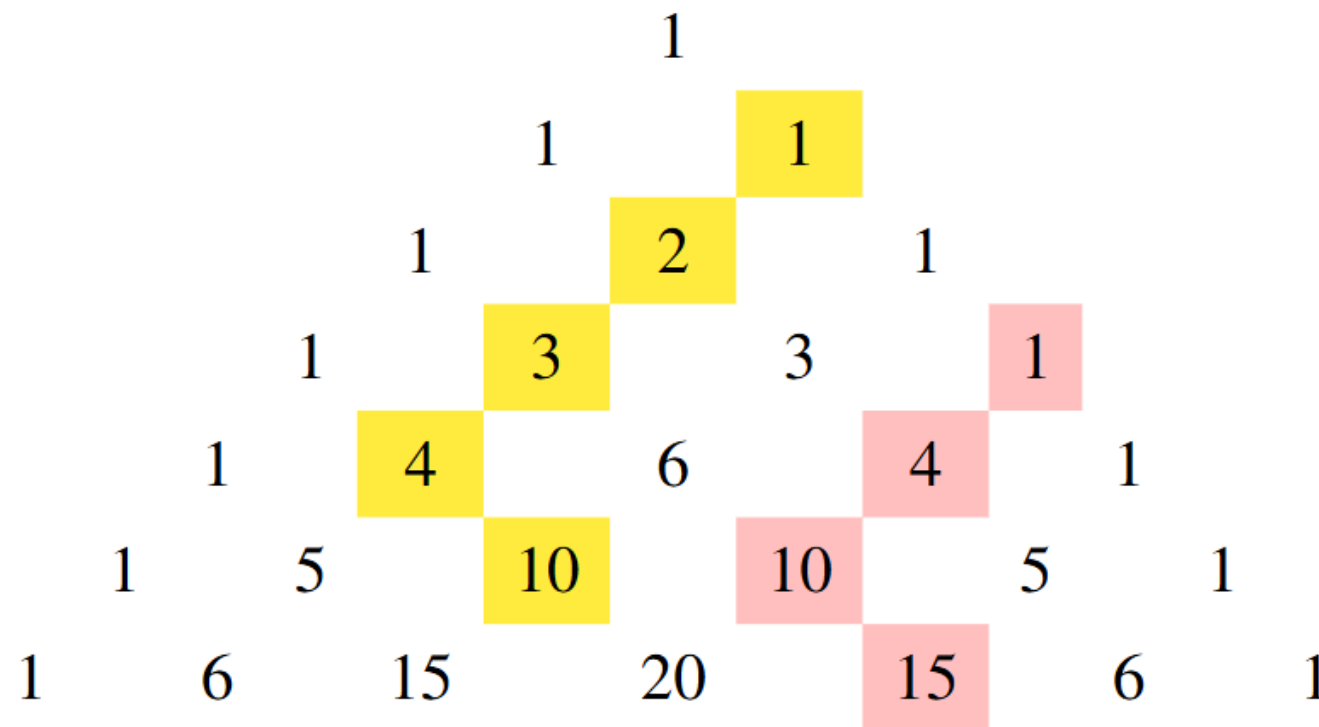
Pascal's triangle (Alternate way to present the table of binomial coefficients)

- **Initial conditions:** Each row starts with $\binom{n}{0} = 1$ and ends with $\binom{n}{n} = 1$.
- **Recursion:** For the rest, each entry is the sum of the two numbers it's in-between on the row above. E.g. $6 + 4 = 10$.



Diagonal sums

- Form a diagonal from a 1 on the right edge, in direction $\swarrow$ as shown, for any number of cells, and then turn $\searrow$ for one cell.



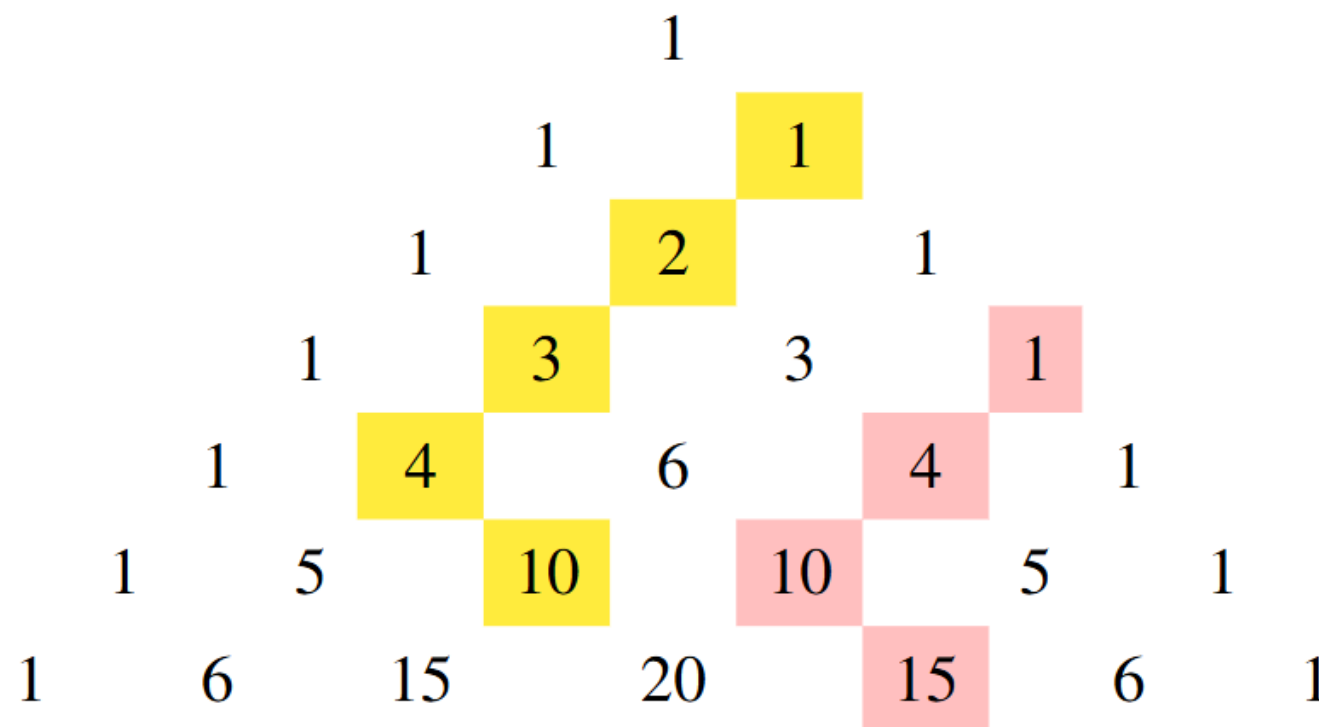
Diagonal sums

- Yellow: $1 + 2 + 3 + 4 = 10$

$$\binom{1}{1} + \binom{2}{1} + \binom{3}{1} + \binom{5}{1} = \binom{5}{2}$$

- Pink: $1 + 4 + 10 = 15$

$$\binom{3}{3} + \binom{4}{3} + \binom{5}{3} = \binom{6}{4}$$

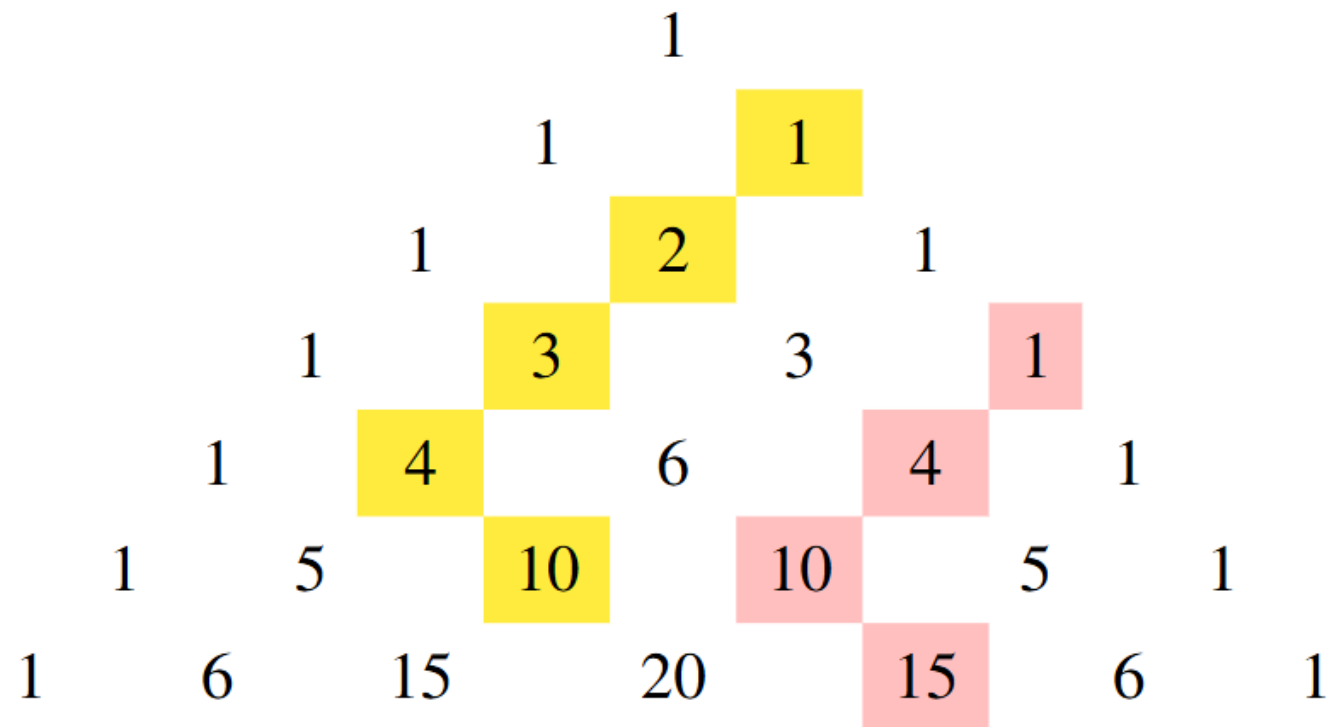


Diagonal sums

We formal a pattern:

$$\binom{k}{k} + \binom{k+1}{k} + \binom{k+2}{k}$$

$$+ \dots + \binom{n}{k} = \binom{n+1}{k+1}$$



Diagonal sums

Theorem 6. For integers $0 \leq k \leq n$:

$$\binom{k}{k} + \binom{k+1}{k} + \binom{k+2}{k} + \cdots + \binom{n}{k} = \binom{n+1}{k+1}$$

Diagonal sums

Theorem 6. For integers $0 \leq k \leq n$:

$$\binom{k}{k} + \binom{k+1}{k} + \binom{k+2}{k} + \cdots + \binom{n}{k} = \binom{n+1}{k+1}$$

Proof. Prove by counting $(k+1)$ -element subsets of $[n+1]$ in two ways.

- First way: The number of such subsets is $\binom{n+1}{k+1}$
- Second way: Categorize subsets by their largest element. (**Left as home exercise**).

Special cases of $(1 + x)^n$ **Theorem 7.**

$$\begin{aligned}(1 + x)^n &= \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{k}x^k + \cdots + \binom{n}{n}x^n \\ &= \sum_{k=0}^n \binom{n}{k} x^k 1^{n-k} = \sum_{k=0}^n \binom{n}{k} x^k\end{aligned}$$

Special cases of $(1 + x)^n$

We will give combinatorial interpretations of these special cases:

For $n \geq 0$,

$$2^n = (1 + 1)^n = \sum_{k=0}^n \binom{n}{k}$$

For $n \geq 0$,

$$0^n = (-1 + 1)^n = \sum_{k=0}^n \binom{n}{k} (-1)^k$$

For $n \geq 0$,

$$3^n = (2 + 1)^n = \sum_{k=0}^n \binom{n}{k} 2^k$$

Example 30

Show that:

$$\binom{n}{k} \binom{k}{m} = \binom{n}{m} \binom{n-m}{k-m}$$

Example 30

Show that:

$$\binom{n}{k} \binom{k}{m} = \binom{n}{m} \binom{n-m}{k-m}$$

Solution. The left-hand side counts the ways to select a group of k people chosen from a set of n people, and then to select a subset of m leaders within the group of k people.

The right-hand side, first select the subset of m leaders from the set of n people and then select the remaining $k - m$ members of the group from the remaining $n - m$ people.

Well-known binomial identities

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n} = 2^n \quad (1)$$

$$\binom{n}{0} + \binom{n+1}{1} + \binom{n+2}{2} + \cdots + \binom{n+r}{r} = \binom{n+r+1}{r} \quad (2)$$

$$\binom{r}{r} + \binom{r+1}{r} + \binom{r+2}{r} + \cdots + \binom{n}{r} = \binom{n+1}{r+1} \quad (3)$$

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \binom{n}{2}^2 + \cdots + \binom{n}{n}^2 = \binom{2n}{n} \quad (4)$$

Well-known binomial identities

$$\sum_{k=0}^m \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r} \quad (5)$$

$$\sum_{k=0}^m \binom{m}{k} \binom{n}{r+k} = \binom{m+n}{m+r} \quad (6)$$

$$\sum_{k=0}^m \binom{m-k}{k} \binom{n+k}{s} = \binom{m+n+1}{r+s+1} \quad (7)$$

Example 31

Evaluate the sum

$$1 \times 2 \times 3 + 2 \times 3 \times 4 + \cdots (n - 2)(n - 1)n.$$

Example 31

Evaluate the sum

$$1 \times 2 \times 3 + 2 \times 3 \times 4 + \cdots (n - 2)(n - 1)n.$$

Solution. The general term in this sum:

$$(k - 2)(k - 1)k = \frac{k!}{(k - 3)!} = P(k, 3).$$

Example 31

Furthermore, we have:

$$C(k, 3) = \frac{k!}{(k-3)! 3!} = \frac{P(k, 3)}{3!}.$$

So, the given sum can be re-written as:

$$3! \binom{3}{3} + 3! \binom{4}{3} + \cdots + 3! \binom{n}{3} = 3! \left\{ \binom{3}{3} + \binom{4}{3} + \cdots + \binom{n}{3} \right\} = 3! \binom{n+1}{4}.$$

Example 32

Evaluate the sum

$$1^2 + 2^2 + 3^2 + \cdots + n^2.$$

Example 32

Evaluate the sum

$$1^2 + 2^2 + 3^2 + \cdots + n^2.$$

Solution. In the general term of this sum, it can be written as:

$$k^2 = k(k - 1) + k,$$

Example 32

So, the given sum can be written as:

$$\begin{aligned}
 & [(1 \times 0) + 1] + [(2 \times 1) + 2] + [(3 \times 2) + 3] + \cdots + [n(n-1) + n] \\
 &= [(2 \times 1) + (3 \times 2) + \cdots + n(n-1)] + [1 + 2 + 3 + \cdots + n] \\
 &= \left[2 \binom{2}{2} + 2 \binom{3}{2} + \cdots + 2 \binom{n}{2} \right] + \left[\binom{1}{1} + \binom{2}{1} + \cdots + \binom{n}{1} \right] \\
 &= 2 \binom{n+1}{3} + \binom{n+1}{2}
 \end{aligned}$$

Q & A